Project Documentation

Project title: **Script-Controlled ACL – Restrict Record Access Based on Field Value**

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ABSTRACT

Access Control Lists (ACLs) are an important security mechanism in ServiceNow used to control who can access, create, modify, or delete records. This project focuses on implementing a **script-controlled ACL mechanism to restrict record access based on a field value**.

The project is developed using the ServiceNow Developer Instance. A student/institution details table is created with fields such as Student Roll Number, Student Name, Faculty Name, Branch, Phone Number, and Description. Different custom roles are created to provide different levels of access to the records.

The main objective is to ensure that users do not receive unrestricted access to all records. For example, the Read ACL is configured with the `bb1` role and a condition that restricts access to records where the **Branch is EEE**. A scripted ACL is also used to control access based on the user's role. Separate ACLs are created for Write, Create, and Delete operations using the `bb2` and `bb4` roles.

This project demonstrates how ServiceNow ACLs, roles, conditions, and scripts can be combined to implement **role-based and field-based record security**.

Introduction

Access control is an important part of application security. In an application, different users may require different levels of access.

For example:

- Administrators may have full access.
- Managers may have limited access.
- Normal users may have read-only access.

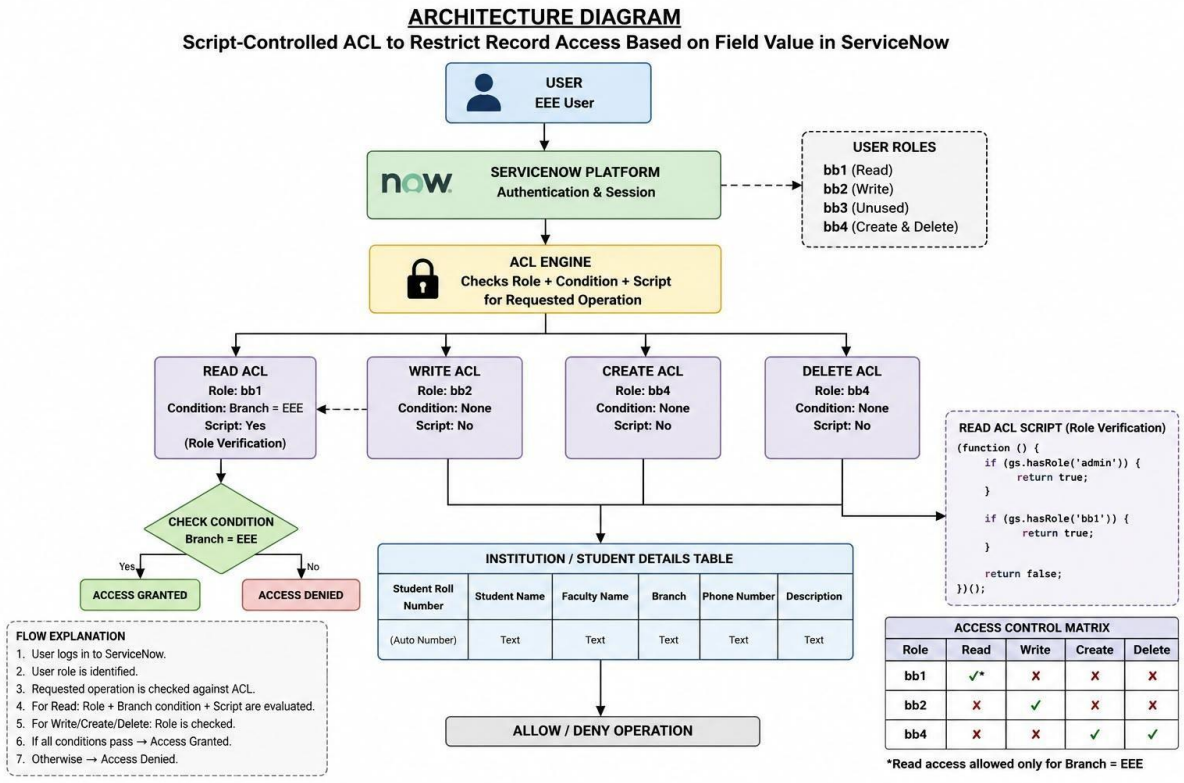
The proposed system uses **ACLs and roles** to implement these permissions. The system allows administrators to create roles, assign permissions, configure ACL rules, and remove unnecessary roles when required.

Objectives

The main objectives of the project are:

- To provide secure access to application resources.
- To create and manage user roles.
- To create and configure Access Control Lists.
- To control read access to resources.
- To assign appropriate permissions to users.
- To prevent unauthorized access.
- To remove unnecessary roles when required

Architecture:



Software Requirements

- **Operating System:** Windows / Linux
- **Platform:** ServiceNow
- **Web Browser:** Google Chrome / Microsoft Edge
- **Internet Connection:** Required
- **Database:** ServiceNow database
- **Development Environment:** ServiceNow Application Navigator and ACL configuration

Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Input:** Keyboard and Mouse
- **Display:** Standard monitor
- **Internet:** Stable internet connection.

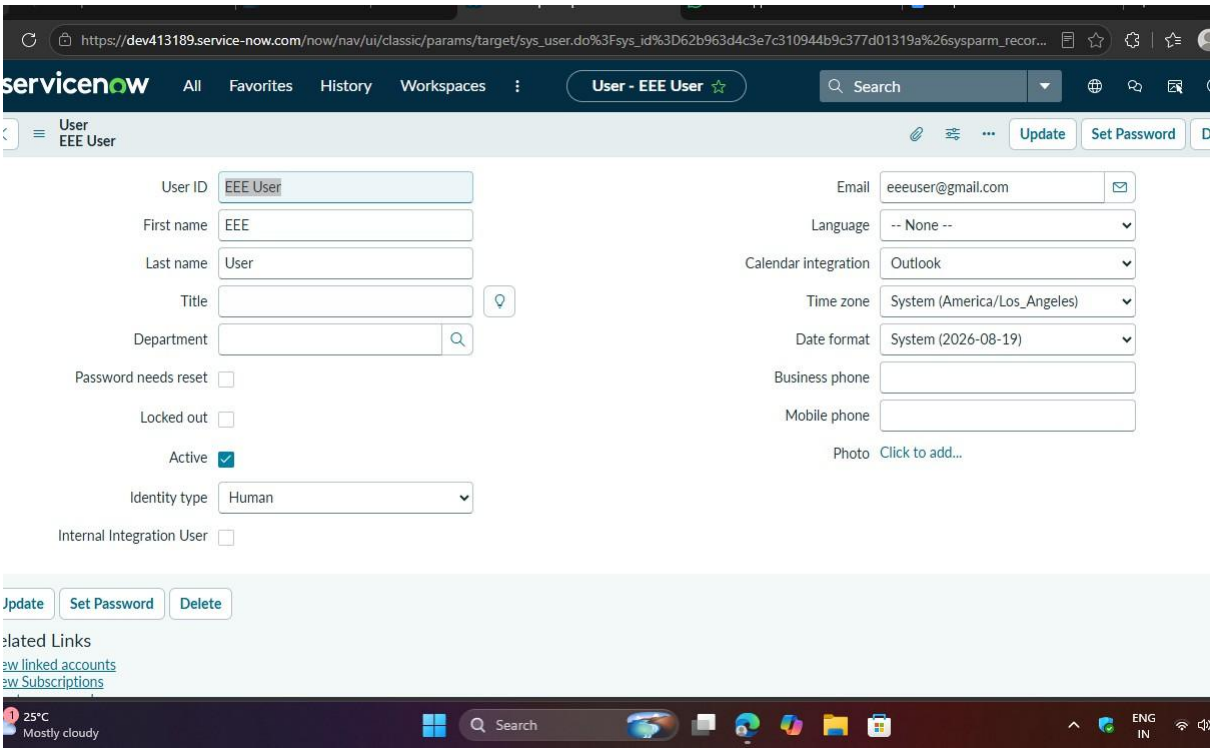
Implementation

The project is implemented using the role and ACL management features of ServiceNow.

Milestone-1 Creation of Users and Roles

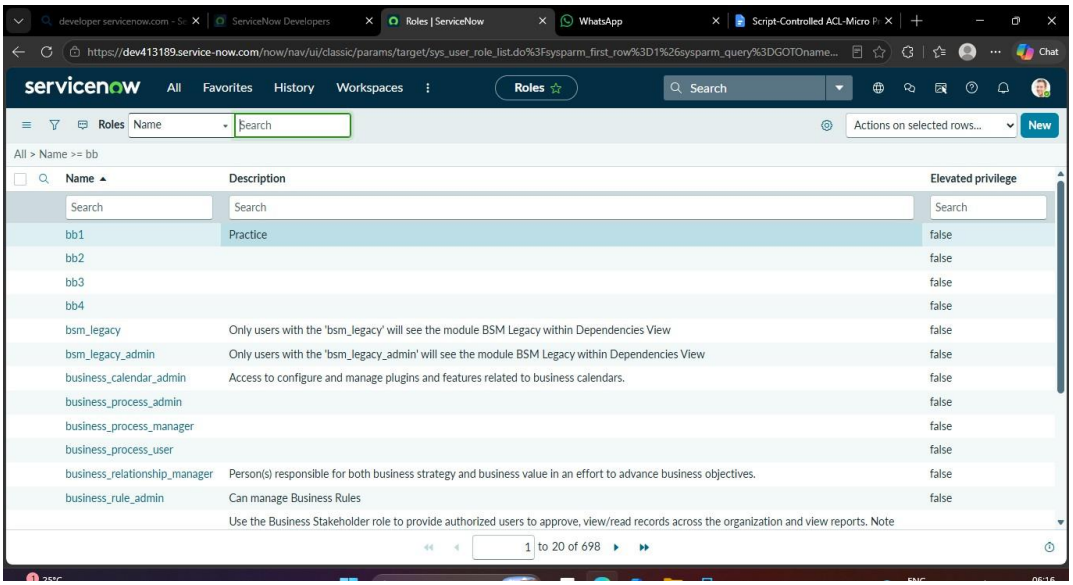
User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
    - Navigate to **User Administration → Users**
    - Click **New**
    - **User ID**: EEE User
    - **First name**: EEE
    - **Last name**: User
    - **Email**: eeeuser@gmail.com
    - Save the record



Role Creation

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User



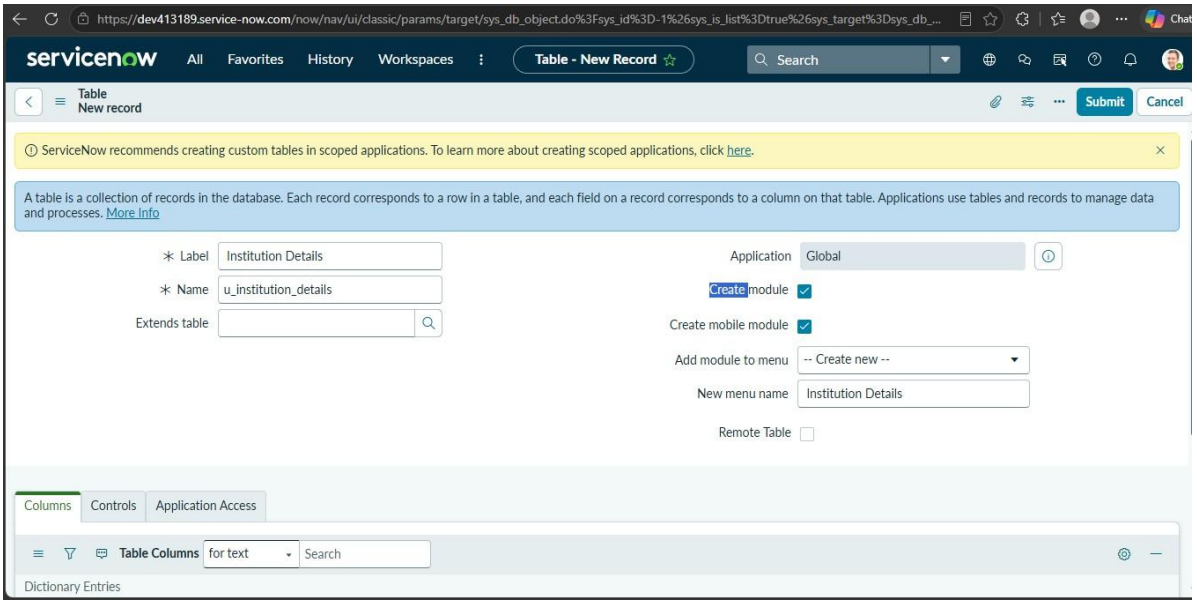
Tables Creation

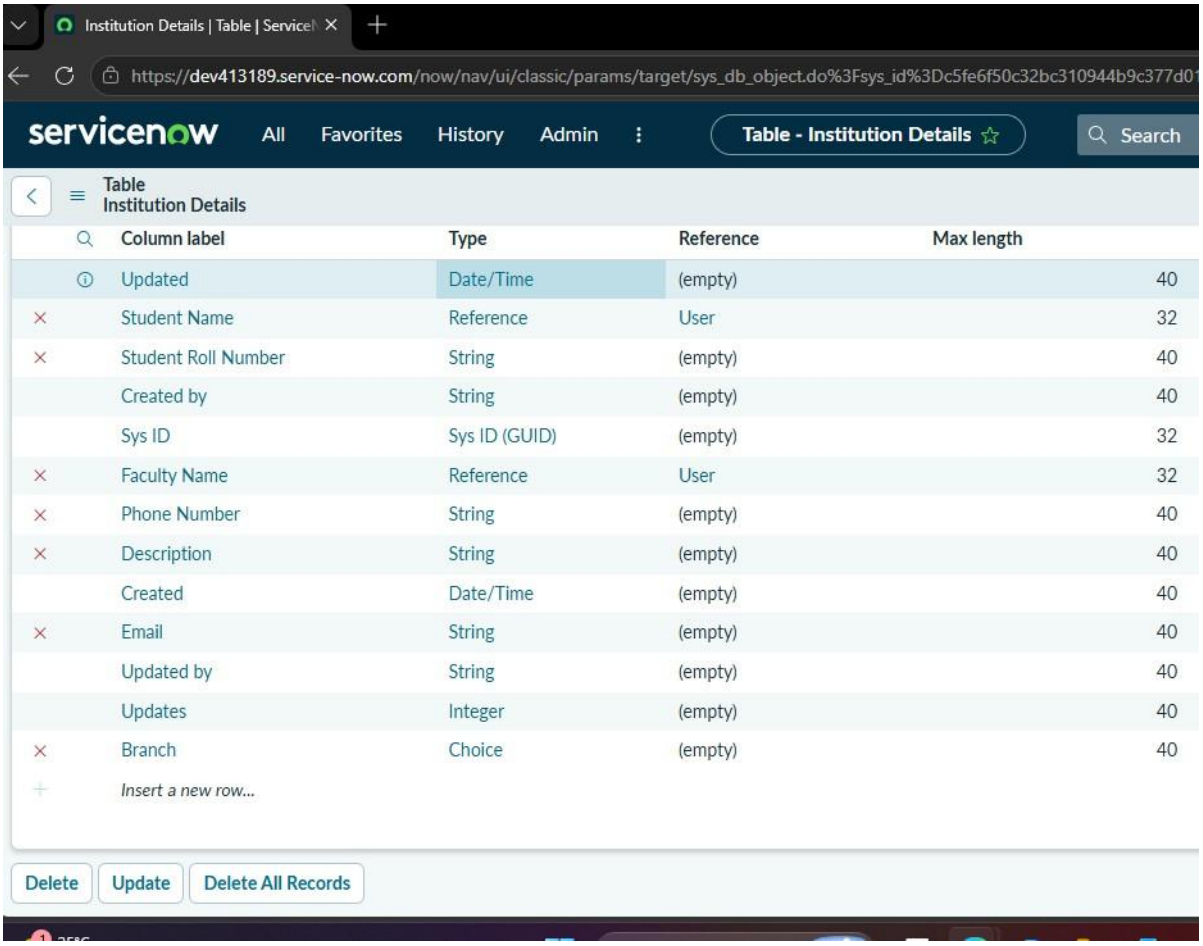
1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
    - **Label:** Institution Details
    - **Name:** u_institution_details
    - **Extends for :** false

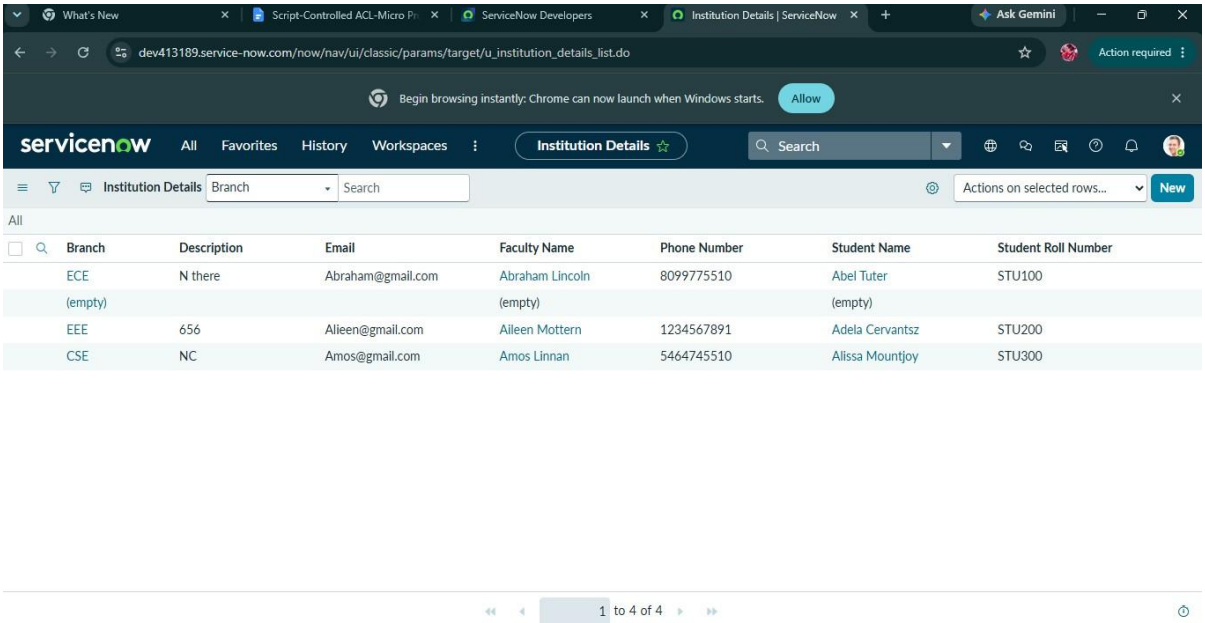
3. .Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
    - **Student Roll Number**– Auto Number
    - **Student Name** – Reference- User
    - **Faculty Name** – Reference- User
    - **Branch** – Choice (ECE, EEE, CSE)
    - **Email**– String
    - **Phone Number**– String
    - **Description**– Multi String

5. **Save the table**

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).




Institution Details | Table | ServiceN
https://dev413189.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3Dc5fe6f50c32bc310944b9c377d01
servicenow
All
Favorites
History
Admin
Table - Institution Details
Search
Table
Institution Details
Column label
Type
Reference
Max length
Updated
Date/Time
(empty)
40
Student Name
Reference
User
32
Student Roll Number
String
(empty)
40
Created by
String
(empty)
40
Sys ID
Sys ID (GUID)
(empty)
32
Faculty Name
Reference
User
32
Phone Number
String
(empty)
40
Description
String
(empty)
40
Created
Date/Time
(empty)
40
Email
String
(empty)
40
Updated by
String
(empty)
40
Updates
Integer
(empty)
40
Branch
Choice
(empty)
40
Insert a new row...
Delete
Update
Delete All Records


What's New
Script-Controlled ACL-Micro Pr
ServiceNow Developers
Institution Details | ServiceNow
Ask Gemini
dev413189.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do
Action required
Begin browsing instantly: Chrome can now launch when Windows starts.
Allow
servicenow
All
Favorites
History
Workspaces
Institution Details
Search
Institution Details
Branch
Search
Actions on selected rows...
New
All
Branch
Description
Email
Faculty Name
Phone Number
Student Name
Student Roll Number
ECE
N there
Abraham@gmail.com
Abraham Lincoln
8099775510
Abel Tuter
STU100
(empty)
(empty)
(empty)
EEE
656
Alieen@gmail.com
Aileen Mottern
1234567891
Adela Cervantsz
STU200
CSE
NC
Amos@gmail.com
Amos Linnan
5464745510
Alissa Mountjoy
STU300
1 to 4 of 4

Creation of Access Control Lists(Read ACL)

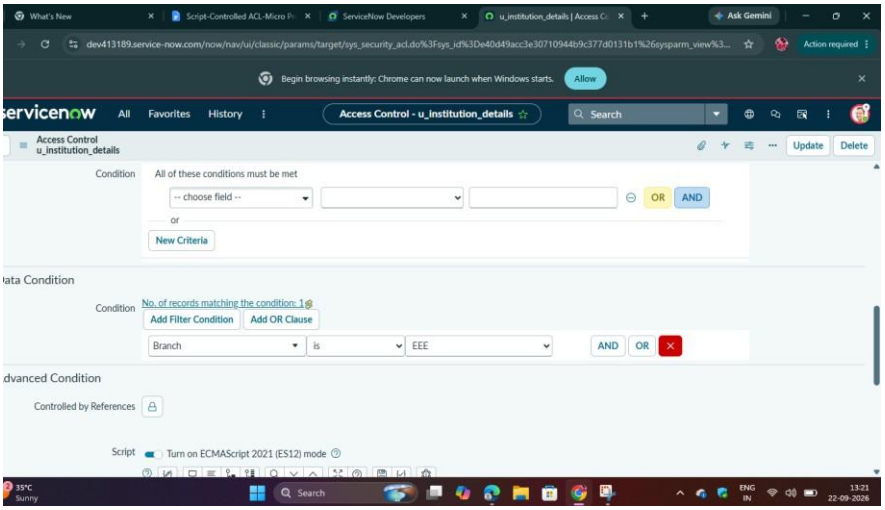
1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role
3. Click **New** to create a new ACL.
4. Enter the following details:
    - **Type:** record
    - **Operation:** read
    - **Name:** u_institution_details
    - **Active:** true

    - **Advanced:** true
5. In the **Requires role** related list, add:
    - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

```
(function () {

   // Allow admin users full access
   if (gs.hasRole('admin')) {
      return true;
   }
   // Allow only EEE branch users to see EEE records
   if (gs.hasRole('bb1')){
      return true;
   }
   // Deny access for all others
   return false;
})();
```

8. Click **Submit** to save the ACL.



Creation of Access Control Lists(Create ACL)

1. **Click** New **to create a new ACL (Create)**
2. Enter the following details:
    - **Type:** record
    - **Operation:** create
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

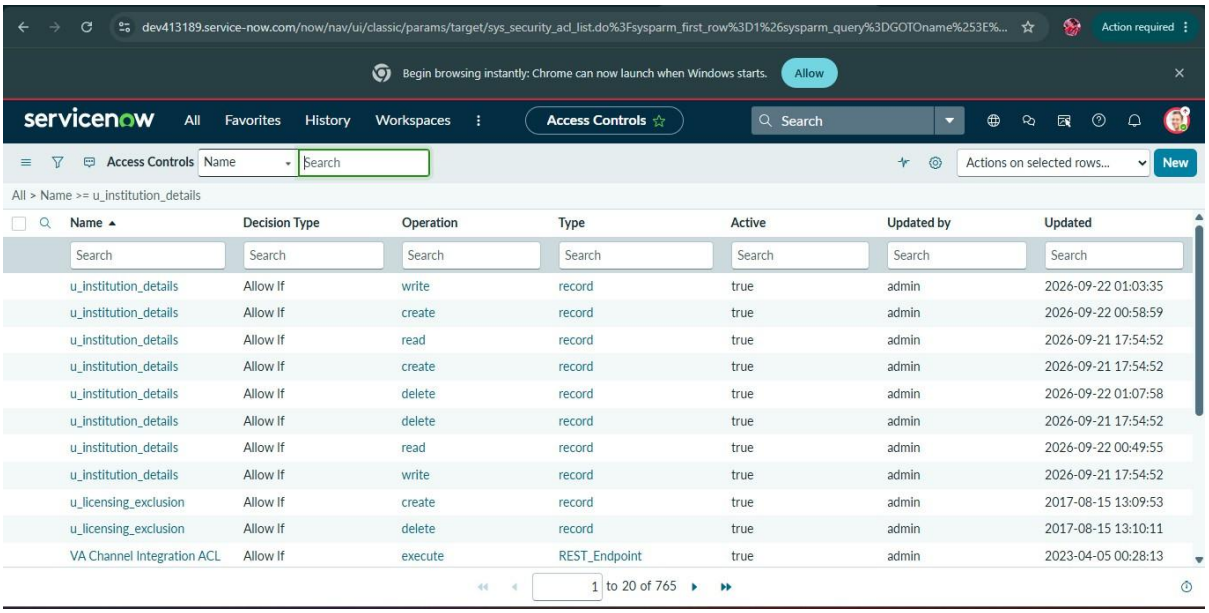
Creation of Access Control Lists(Create ACL)

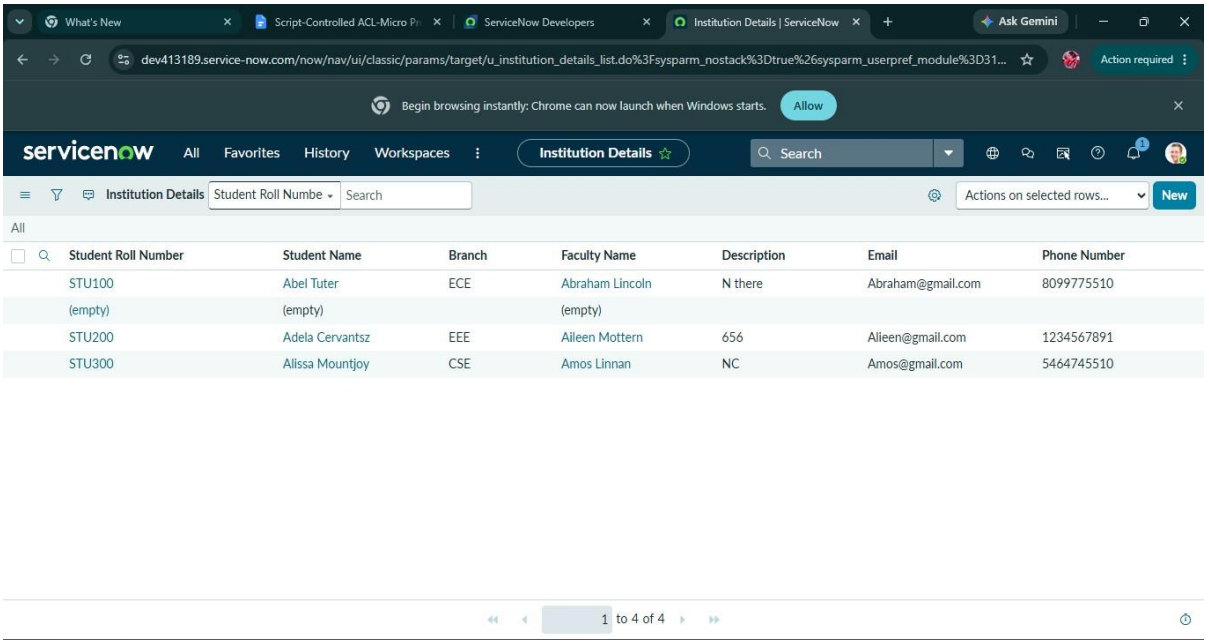
1. **Click** New **to create a new ACL(Create)**
2. Enter the following details:
    - **Type:** record
    - **Operation:** create
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

Creation of Access Control Lists(Delete ACL)

1. **Click** New **to create a new ACL(Delete)**
2. Enter the following details:
    - **Type:** record
    - **Operation:** Delete
    - **Name:** u_institution_details
    - **Active:** true
3. In the **Requires role** related list, add:
    - bb4 (custom role)
4. No need to add any Data condition

Save the ACL Record





Conclusion

The Access Control List Management System provides a secure and organized method for controlling access to application resources. By using roles, permissions, and ACLs, the system allows authorized users to access required resources while restricting unauthorized users. The implementation improves application security and makes user access easier to manage.